Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Rinal Shah, Varun Hotani, Aadi Mehta, Pratik Agrawal
ECE501 Digital Image Processing, Group 15 (Project 17 [VERIFY: confirm project no.])
[Institution / Department – TO FILL]
[Roll numbers / e-mail addresses – TO FILL]

Abstract—[TEAM TO WRITE: summarise problem, method, 13-image scope, principal patch results, weak pixel localization, next step. Do not quote success figures without the pixel IoU.]

Index Terms—fabric defect detection, texture analysis, GLCM, LBP, Mahalanobis distance, adaptive thresholding, AFID

I. Introduction

[TEAM TO WRITE: inspection problem and texture variation; AFID dataset [1], [2]; GLCM [3] and LBP [4]; project objective; scope of this mid-semester report.]

II. Methodology

[TEAM TO WRITE: describe the pipeline in your own words: median filter, CLAHE, patch extraction, GLCM + multiscale LBP descriptor, per-fabric normal model, deviation map, threshold, morphology.] [VERIFY: patch size/stride, median kernel, CLAHE clip/grid, GLCM and LBP settings, k, λ, morphology kernel, minimum area, patch ground-truth rule vs. upgraded code]

$$d_f(\mathbf{x}) = \sqrt{(\mathbf{x}-\boldsymbol{\mu}_f)^\top \boldsymbol{\Sigma}_{f,\mathrm{reg}}^{+} (\mathbf{x}-\boldsymbol{\mu}_f)} \tag{1}$$

$$\boldsymbol{\Sigma}_{f,\mathrm{reg}} = \boldsymbol{\Sigma}_f + \lambda \frac{\mathrm{tr}(\boldsymbol{\Sigma}_f)}{D} \mathbf{I} \tag{2}$$

$$\tau_f = \mu_{d,f} + k\,\sigma_{d,f} \tag{3}$$

[TEAM TO WRITE: define $\mathbf{x}$, $\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_{f,\mathrm{reg}}$, λ, k and the pseudoinverse; state which images fit each reference and how the threshold is set.]

[TEAM TO WRITE: evaluation protocol: patch and pixel definitions of TP/FP/FN/TN, metric formulas, per-image averaging.] [VERIFY: reference-training images, test selection, validation, ROI handling]

III. Results

[TEAM TO WRITE: describe the 13-image evaluation subset, the averaging rule, and the main patterns in Table I.]

IV. Discussions

[TEAM TO WRITE: interpret low pixel precision/IoU, fabric-to-fabric variability, coarse patch boundaries, and what is not yet evaluated.]

[FIGURE PLACEHOLDER: AFID image, ground-truth mask, deviation map, predicted mask — provenance to be confirmed]

Fig. 1. [TEAM: caption, sample filename, and "real AFID" provenance]

V. Conclusion

[TEAM TO WRITE: demonstrated capability, limitations, planned next phase.]

AI assistance: Used to help with coding and debugging.

References

[1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," AUTEX Res. J., vol. 19, no. 4, pp. 363–374, 2019.
[2] AITEX, "AITEX fabric image database (AFID)," https://www.aitex.es/afid/, accessed: Oct. 11, 2026.
[3] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," IEEE Trans. Syst., Man, Cybern., vol. SMC-3, no. 6, pp. 610–621, Nov. 1973.
[4] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," IEEE Trans. Pattern Anal. Mach. Intell., vol. 24, no. 7, pp. 971–987, Jul. 2002.

TABLE I
Mean per-image metrics (%) on the 13-image AFID subset. Patch: 256×256 patches; Pixel: full strip. Pixel F1 equals Dice. Fabric rows average the images of that fabric ID (n).

		Patch					Pixel				
Fabric ID	n	Acc	Prec	Rec	F1	IoU	Acc	Prec	Rec	F1	IoU
All (mean)	13	79.83	21.88	41.54	27.13	20.56	80.66	0.36	43.57	0.72	0.36
00	2	95.55	66.67	100.00	80.00	66.67	89.75	1.16	100.00	2.29	1.16
01	2	74.19	11.11	50.00	18.18	11.11	73.46	0.48	50.00	0.95	0.48
02	2	70.97	11.11	50.00	18.18	10.00	72.27	0.28	50.00	0.55	0.28
03	2	78.26	20.00	20.00	20.00	12.50	90.34	0.34	33.18	0.68	0.34
04	2	74.19	0.00	0.00	0.00	0.00	76.56	0.00	0.00	0.00	0.00
05	1	96.55	66.67	100.00	80.00	66.67	87.53	0.20	100.00	0.40	0.20
06	2	77.42	0.00	0.00	0.00	0.00	78.12	0.00	0.00	0.00	0.00